Test Normal Cases

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Weeks 15 and 23: keep evaluation unfamiliar and report denominators.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Plan routine cases with expected results derived from the requirements. Try the recall before opening notes.

Your learning target

I can plan routine cases with **expected result**s derived from the requirements.

I can compare a fixed prototype and manual **baseline** on the same requests.

I can report every result and explain what a small **test set** cannot establish.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Repeated success on ordinary cases is necessary but not sufficient.



Your target

LEARNING TARGET A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

Learner name:

Readiness for the connected explanation

How does an **expected result** differ from an observed result, and why keep both when they disagree?

This week’s end goal

By the end, I can test routine cases against expectations and count full passes honestly.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Test Normal Cases case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY The sorter is tested on five common classroom items with expected bins written before each run.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

This week you will turn separate paper traces into a test record. Choose several ordinary requests, write the required answers first, then run the same saved prototype on every request.

Testing is useful when it can reveal a mistake, so keep failures as carefully as successes. A demonstration chosen because it works is not a complete evaluation.

Use catalogue B version 1 in order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These fictional values assume materials are available. **Input**s are whole minutes from 0 to 30 and Quiet required Yes or No. Time must fit AND either quiet is not required or the activity is quiet. No allows either kind.

Select the first fit with a correct reason; otherwise report no match. Missing or invalid **input** requires clarification. No booking or publishing takes place.

Keep the P1 version 1 instructions from Week 32: validate and stop with clarification if **input** is invalid; otherwise start at B1, check both conditions, suggest the first passing row and stop.

After a failure advance one row; report no match only after all rows fail. Reset to validation and B1 for each new valid request. The manual **baseline** marks every row against both conditions and selects the first fit.

Both methods must follow the same contract.

STUDENT LESSON · 3 OF 18

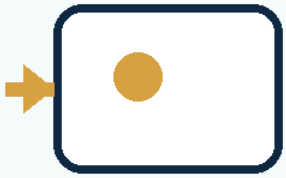
See how it works

> Follow the mechanism for Test Normal Cases. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



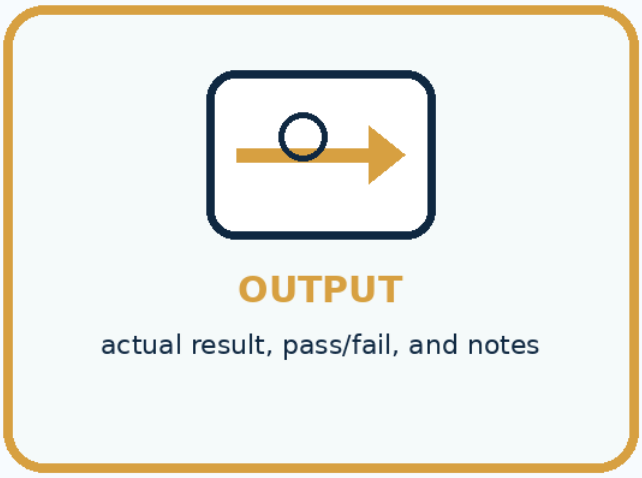
INPUT

planned test input and expected result



PROCESS

run unchanged prototype and observe



INPUT planned test **input** and **expected result**

PROCESS run unchanged prototype and observe

OUTPUT actual result, pass/fail, and notes

Key idea

CORE IDEA Repeated success on ordinary cases is necessary but not sufficient.

How the mechanism works

UNDER THE HOOD A **test case** includes **input**, setup, expected behavior, actual behavior, and result. Precommitting expectations reduces hindsight bias.

Repeated normal tests establish basic reliability but say little about uncommon conditions.

Draw a fresh example with all three stages:

A closer explanation

From a planned case to evidence

BEFORE THE RUN	DURING THE RUN	AFTER THE RUN
Write input and expected result	Use the saved version	Compare with the requirement
State why the case matters	Record output and reason	Mark pass or explain failure
Label both comparison methods	Keep every trace	Report counts and limits

Keep the catalogue, requests and pass criteria the same for the prototype and baseline.

Give each case an ID. Record its exact **input**, purpose, expected **output** and expected explanation before running it. Then add observed **output**, observed explanation, pass or fail, and a trace reference.

Put the prototype version, catalogue version and **baseline** method at the top of the sheet.

A pass needs the correct result and a reason supported by the checked conditions; the right activity with a false reason is not a full pass.

Our routine set R contains four valid requests. R1 is 11 minutes Yes, checking an early quiet match. R2 is 9 minutes No, checking that the search reaches a later **fitting** row.

R3 is 6 minutes No, checking that a valid request can exhaust the list. R4 is 9 minutes Yes, checking that a time fit alone is insufficient. These cases do not cover every accepted **input** or validation error.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

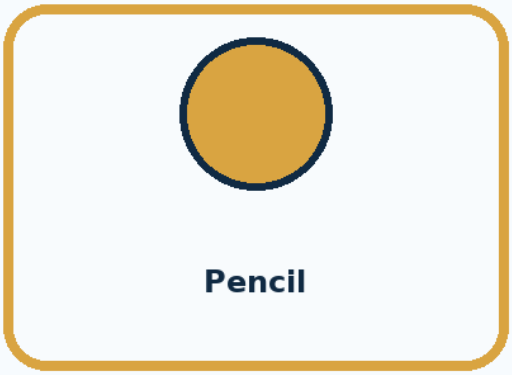
Compare the cases

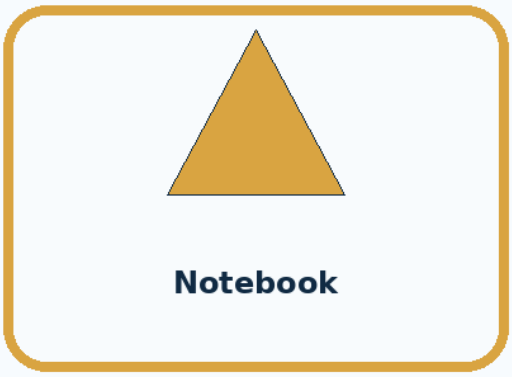
> Compare the supplied cases for Test Normal Cases. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Pencil Clue: common clear item. Meaning: normal case.

The case to investigate

CASE 2 — Notebook Clue: common clear item. Meaning: normal case.

The case to investigate

CASE 3 — Expected written later Clue: changed after result. Meaning: weak **evidence**.

The case to investigate

CASE 4 — Same **process** each run Clue: controlled setup. Meaning: **reproducible**.

Keep this boundary clear

BOUNDARY Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Prototype reports none for request 6 instead of P1.

Predicted result

Prediction: That run still fails.

Why the result follows

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Test Normal Cases to explain a concrete decision, rather than only memorising definitions.

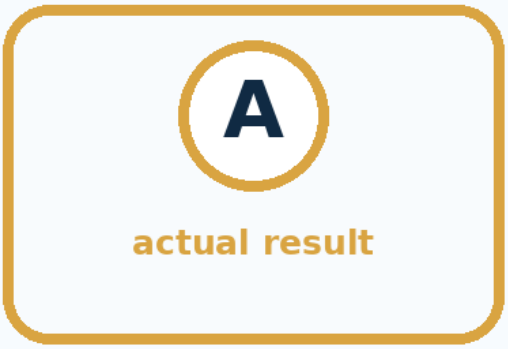
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
test case	planned input and expected behavior	
expected result	prediction written before running	

Word	Meaning	My example
actual result	observed behavior	
reproducible	another tester can repeat the setup	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR Five passes prove the system is ready for every item.

Use one vocabulary word and one piece of **evidence** to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Test set	A collection of cases used together to check a version
Pass criterion	The requirement an observed result must meet to count as a pass
Coverage	Which conditions or behaviours the selected tests examine
Baseline	A simple comparison method tested under the same conditions

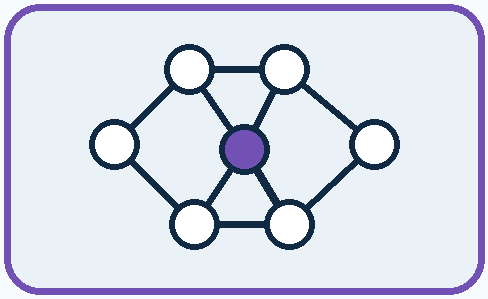
Follow a worked example

> Follow the worked trace for Test Normal Cases, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE The sorter is tested on five common classroom items with expected bins written before each run.

Step 1

STEP 1 — OBSERVE planned test **input** and **expected result**

Step 2

STEP 2 — TRACE run unchanged prototype and observe

Step 3

STEP 3 — CONCLUDE **actual result**, pass/fail, and notes

Model answer

MODEL REASONING Write the expected **output** first, keep the prototype version fixed, record every actual **output**, and count passes without deleting failures.

Explain that five normal successes are necessary **evidence**, not full proof.

A claim to avoid

DO NOT OVERCLAIM Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

What new **evidence** could change this answer?

A closer explanation

Case and request	Expected output and reason	Fictional observed Q version 0
R1 11 minutes Yes	B1; 10 fits 11 and B1 is quiet	B1 with the same correct reason; pass
R2 9 minutes No	B2; 8 fits 9, quiet unrestricted, B1 too long	B2 with the same correct reason; pass
R3 6 minutes No	No match; all activities take longer	No match with the same correct reason; pass
R4 9 minutes Yes	No match; B2 is not quiet and other rows take longer	B2 because 8 fits 9; fail

Worked explanation · Test Normal Cases

This is an invented record for a comparison prototype Q version 0, not a claim about your P1 runs. Q ignores the quiet condition when searching. Its trace explains the R4 failure.

In the same fictional comparison, the manual **baseline** produced the **expected result** and correct explanation on all four cases: 4 of 4 full passes. Q achieved 3 of 4. The denominator includes the failed case.

Worked explanation · Test Normal Cases

On this set Q has fewer full passes than the **baseline**. This supports investigating its quiet check; it does not establish performance on all requests. Do not claim a speed advantage because we did not measure time.

Correct implementations of P1 and the **baseline** can both pass all four, in which case these results show no correctness advantage for either method.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Test Normal Cases

A fictional pencil-pouch finder has been run on four ordinary requests.

The facts we will use

Supplied facts

Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9.

Supplied facts

Requests 2,4,6,8; recorded **output**s P1,P2,P1,none. These are invented records, not class results.

AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/33

Test Normal Cases

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write expectations

For 2,4,6,8 expect P1,P2,P2,none.



2. Compare outputs

Runs at 2,4 and 8 agree; run at 6 disagrees.



3. Trace failure

P1 holds only 3, so it cannot satisfy request 6.



4. Report

Three of four routine cases pass; retain the failure.

CHANGE ONE THING

Prototype reports none for request 6 instead of P1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write expectations	For 2,4,6,8 expect P1,P2,P2,none.
2. Compare outputs	Runs at 2,4 and 8 agree; run at 6 disagrees.
3. Trace failure	P1 holds only 3, so it cannot satisfy request 6.
4. Report	Three of four routine cases pass; retain the failure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

What this example does not establish

Keep the stated limit

Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

Keep the stated limit

Read the labelled picture: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Keep the stated limit

Change one thing: Prototype reports none for request 6 instead of P1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Test Normal Cases claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Pencil	common clear item	normal case

Case	Evidence	What it shows
Notebook	common clear item	normal case
Expected written later	changed after result	weak evidence
Same process each run	controlled setup	reproducible

Repair two tempting claims

Challenge claim A

CLAIM A Five passes prove the system is ready for every item.

Repair:

Challenge claim B

CLAIM B A failed test should be removed because it lowers the score.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A useful report names the version, catalogue, case set, full-pass count and total, then describes each failure and the requirement it breaks. Keep the trace so someone can reproduce the result.

Say which conditions have not been tested. Four passing runs support a claim about those four runs; they do not prove that every possible request works.

Check the claim against the facts

For catalogue B, any request allowing B3 also allows the earlier quiet B1; any request allowing B4 also allows earlier B1 when quiet is not required. Therefore the first-match rule never selects B3 or B4 with this catalogue.

Checking those rows in a no-match trace is different from testing a successful **output** from a later row.

Next week we will deliberately examine boundaries and labelled catalogue variations while keeping the requirements explicit.

A team reports "Our prototype passed 4 of 4 routine cases, so it always works and is faster than the baseline." Rewrite this as a claim supported by the evidence, and identify two further checks that would address different gaps.

STUDENT LESSON · 8 OF 18

Practise together

Practise Test Normal Cases with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Repeated success on ordinary cases is necessary but not sufficient.

Use a fresh example to explain **test case**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A failed test should be removed because it lowers the score.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Before testing your own cards, finish all four **expected result**s and agree the **pass criterion**. Swap roles: one pupil operates and the other records.

Keep the same version throughout the set and begin a fresh run for each request. Record what happened even if a card is ambiguous or the operator makes a mistake.

A run that stops without a required answer is a failed run to investigate, not a row to delete.

Run the manual **baseline** on the same four requests and catalogue. Record its **actual result**s too; a person can make an error in either method.

If you find a defect, preserve this set's results, save a new version and start a separately labelled set after the repair. Never combine the best result from each version into a score for one version.

In the fictional Q record, give the full-pass fraction for each method and explain the R4 failure. A teammate removes R4 and reports Q as 3 of 3. Explain how that changes the question being answered. Then state one condition that set R has not checked.

Practise with less help: Run a fair comparison

Use workbook W2 for the connected example “Testing routine cases”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

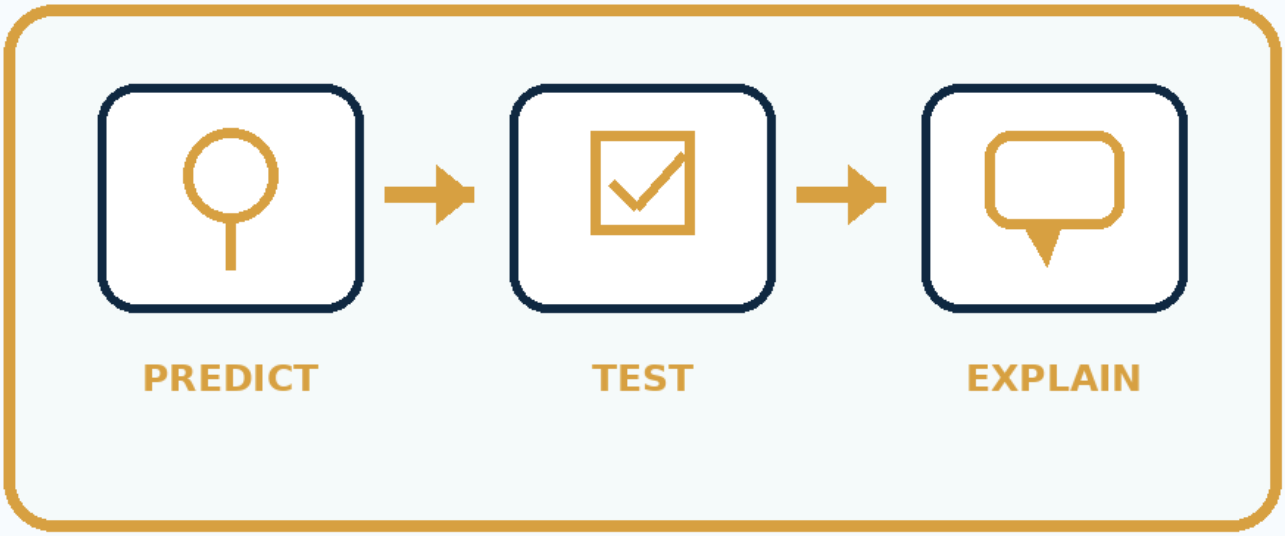
Run the investigation

> Investigate Test Normal Cases. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Five-Case Normal Test

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Run five planned tests and log expected versus actual.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

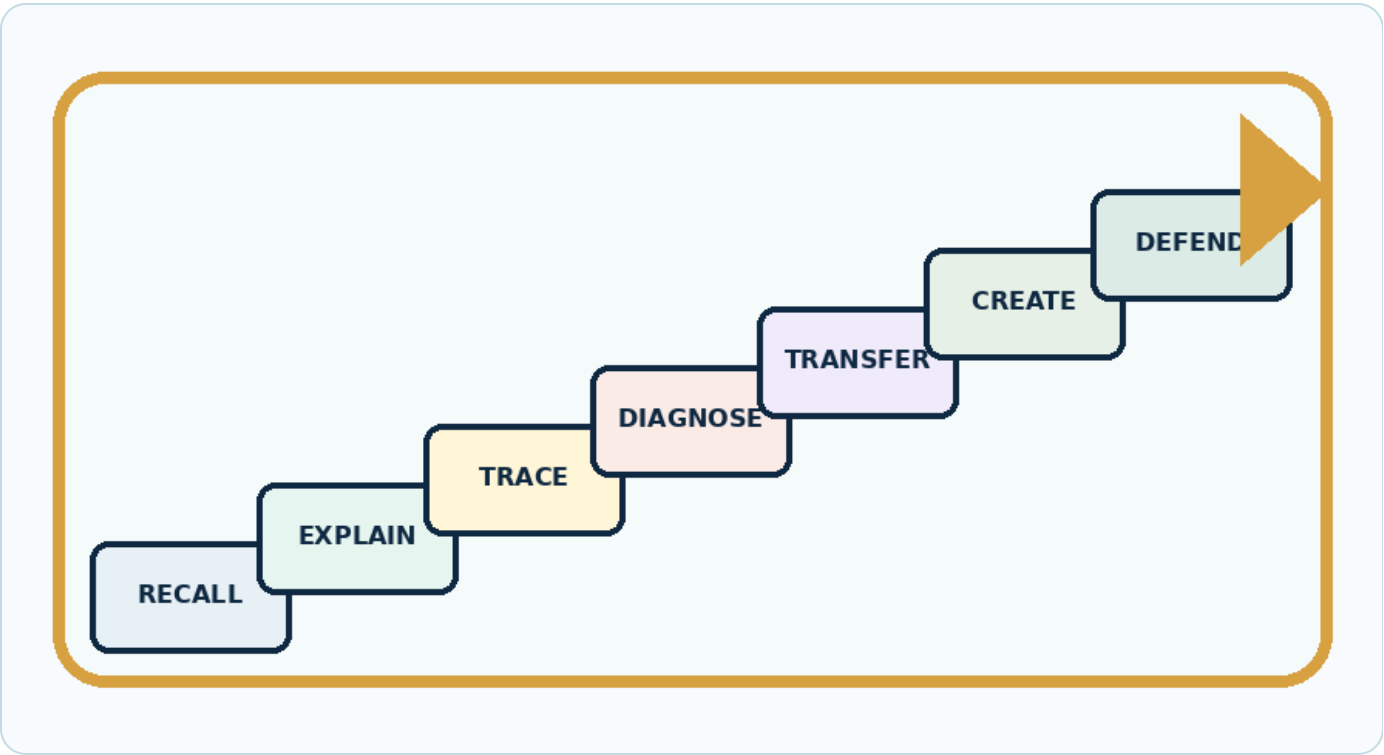
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Test Normal Cases. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **test case**.

EXPLAIN Repeated success on ordinary cases is necessary but not sufficient.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Five passes prove the system is ready for every item.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

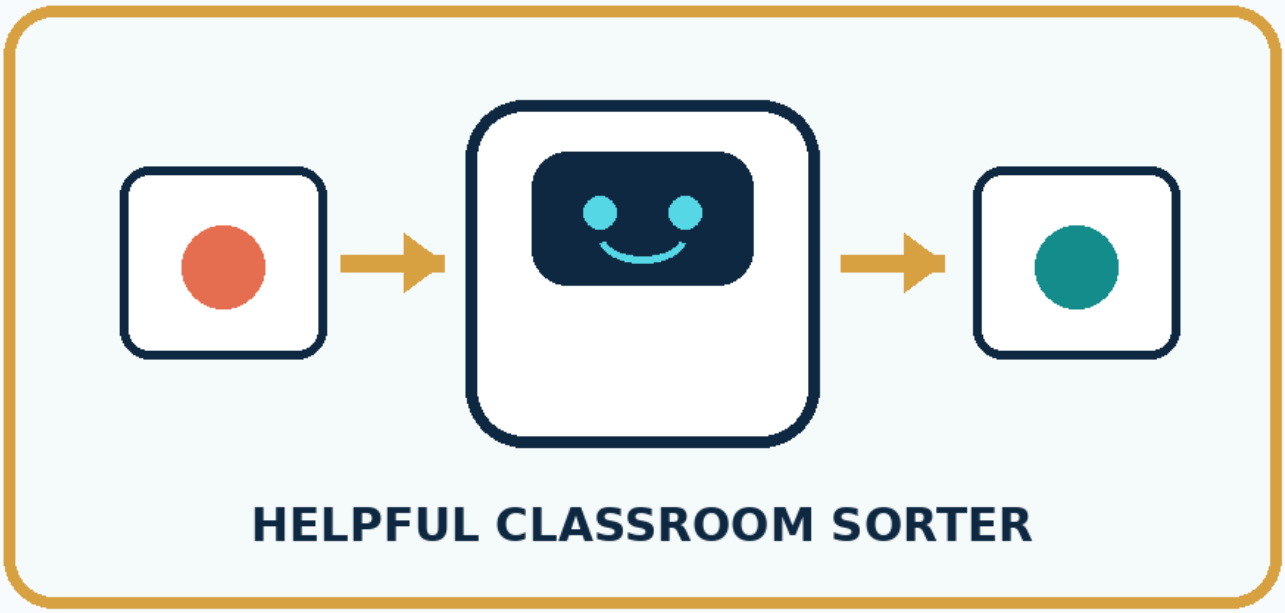
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



> THIS WEEK'S CHECKPOINT Create and run the sorter's normal-case test plan with unique IDs and version label.

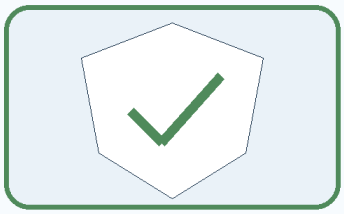
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not change the prototype halfway through a **test set** and combine results as if one version was tested.

My project decision:

Evidence for your project

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Run five planned tests and log expected versus actual before calculating

the pass count.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Write the expected **output** first, keep the prototype version fixed, record every actual **output**, and count passes without deleting failures. Explain that five normal successes are necessary **evidence**, not full proof.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Test Edge Cases and Fairness.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Test Normal Cases case for supported practice. Repeated examples do not become fresh assessments.

Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Four routine cases are chosen before testing. Three are correct.

The question

> What can be reported?

Worked explanation

3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Report the cases actually tested

Evidence or stage	Value or result
Routine cases	3/4

Four routine cases are chosen before testing. Three are correct.

> What can be reported?

3/4 on those cases. Preserve the failure and trace its cause. This does not guarantee unfamiliar-case performance.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit a new test record

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and **expected result**

> **Write or draw your reasoning and evidence.**

Connect to your project

Create and run the sorter's normal-case test plan with unique IDs and version label.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Test Normal Cases, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional pencil-pouch finder has been run on four ordinary requests.

By the end, I can	How I will show it
Plan routine cases with expected results derived from the requirements	Explain the deciding step, show a trace or test, and state one limit.
Compare a fixed prototype and manual baseline on the same requests	Explain the deciding step, show a trace or test, and state one limit.
Report every result and explain what a small test set cannot establish	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Test Normal Cases without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Plan routine cases with **expected result**s derived from the requirements

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional pencil-pouch finder has been run on four ordinary requests.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four routine cases pass; retain the failure.

1. Trace failure

2. Write expectations

3. Compare **output**s

4. Report

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report Three of four routine cases pass; retain the failure. An ordinary case is not automatically correct.

Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Prototype reports none for request 6 instead of P1. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That run still fails.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Repair a missing building block

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?.

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

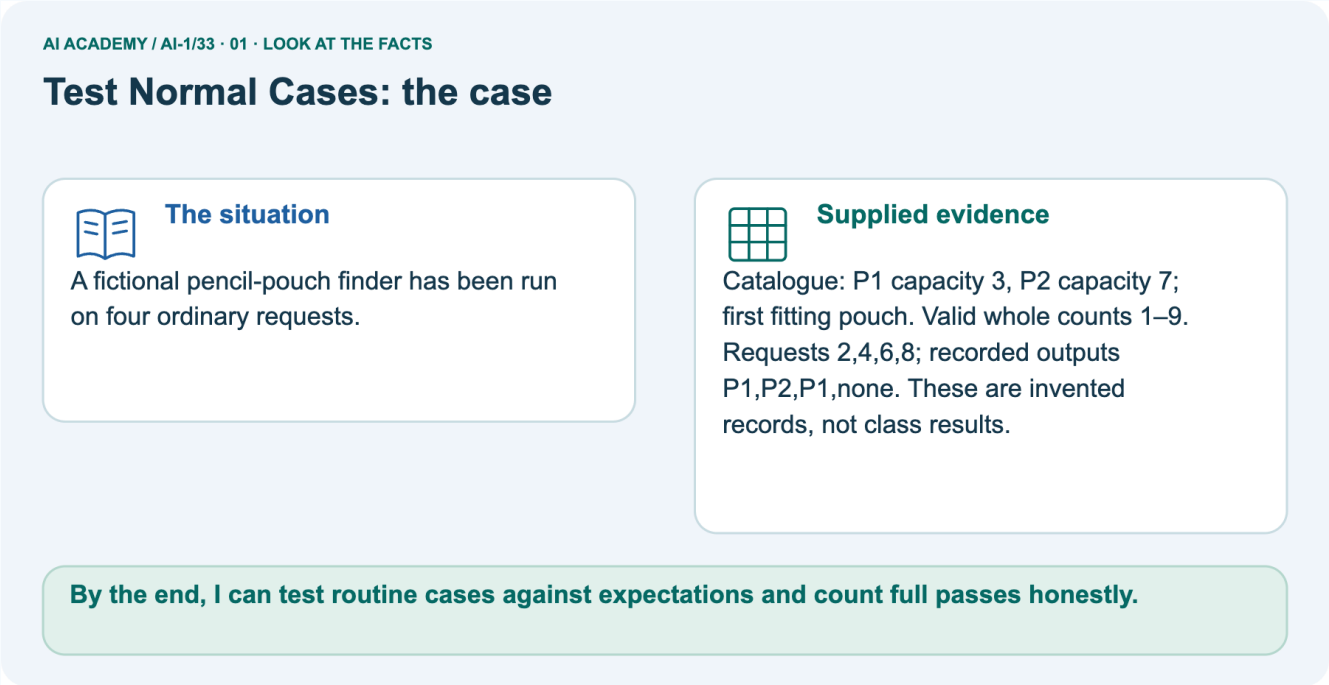
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Test Normal Cases graphic using its labels, deciding evidence and relevant limitation.

01. Test Normal Cases: the case



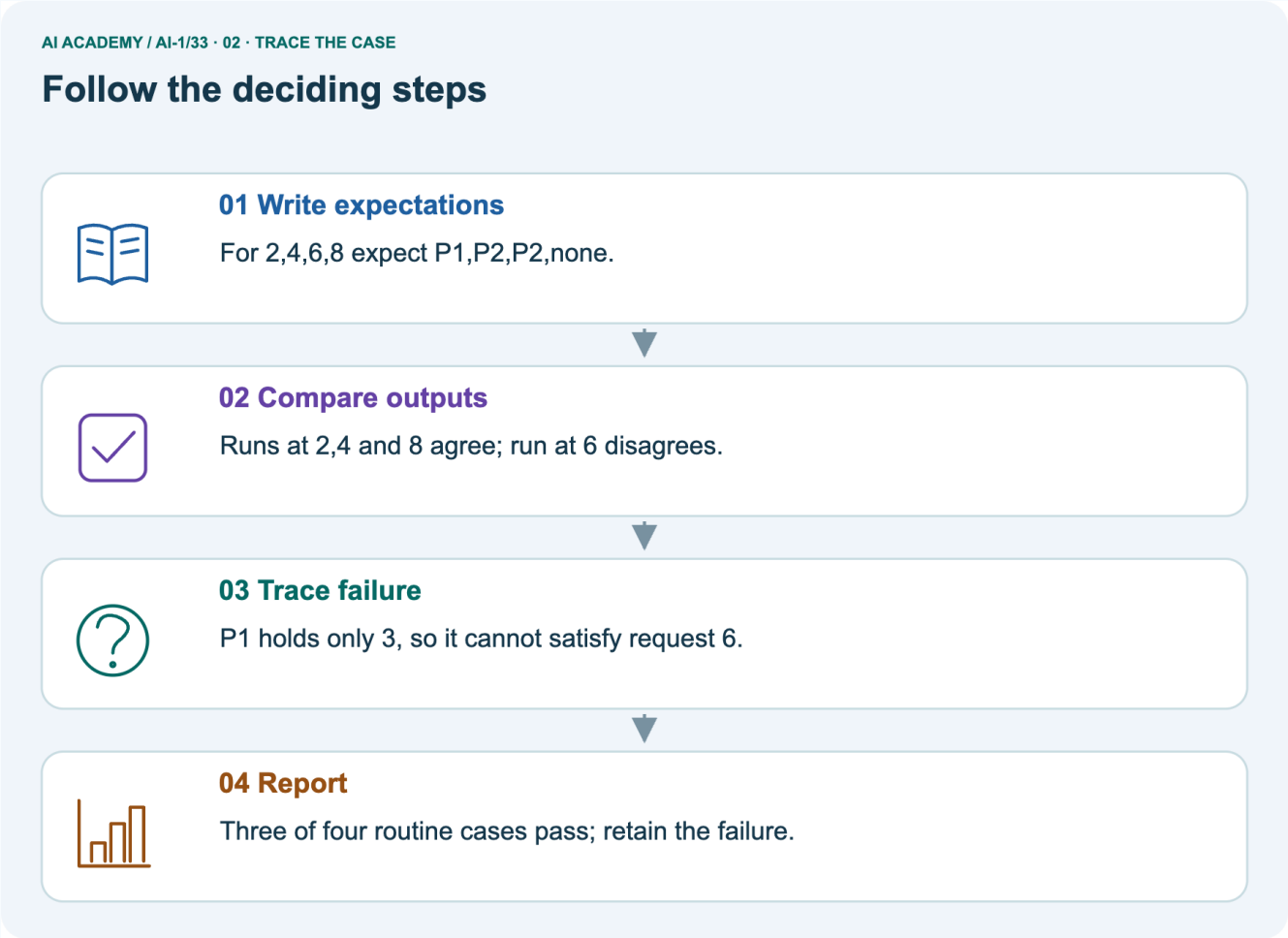
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional pencil-pouch finder has been run on four ordinary requests. Supplied facts: Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9. Requests 2,4,6,8; recorded **output**s P1,P2,P1,none.

These are invented records, not class results.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

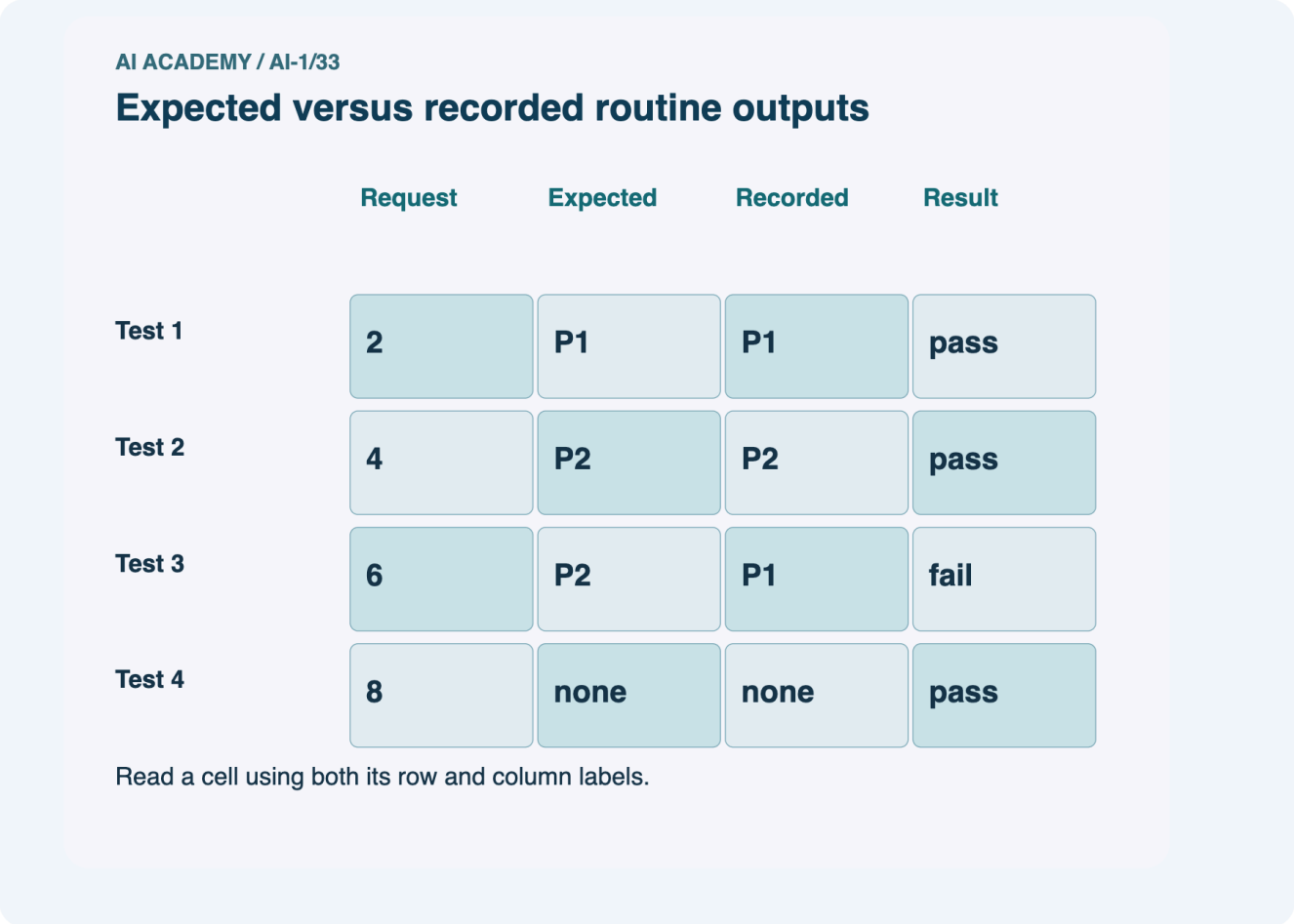


Follow the labelled picture with the matching explanation. Open the vector diagram

Write expectations: For 2,4,6,8 expect P1,P2,P2,none. → Compare **output**s: Runs at 2,4 and 8 agree; run at 6 disagrees. → Trace failure: P1 holds only 3, so it cannot satisfy request 6. → Report: Three of four routine cases pass; retain the failure.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Expected versus recorded routine outputs



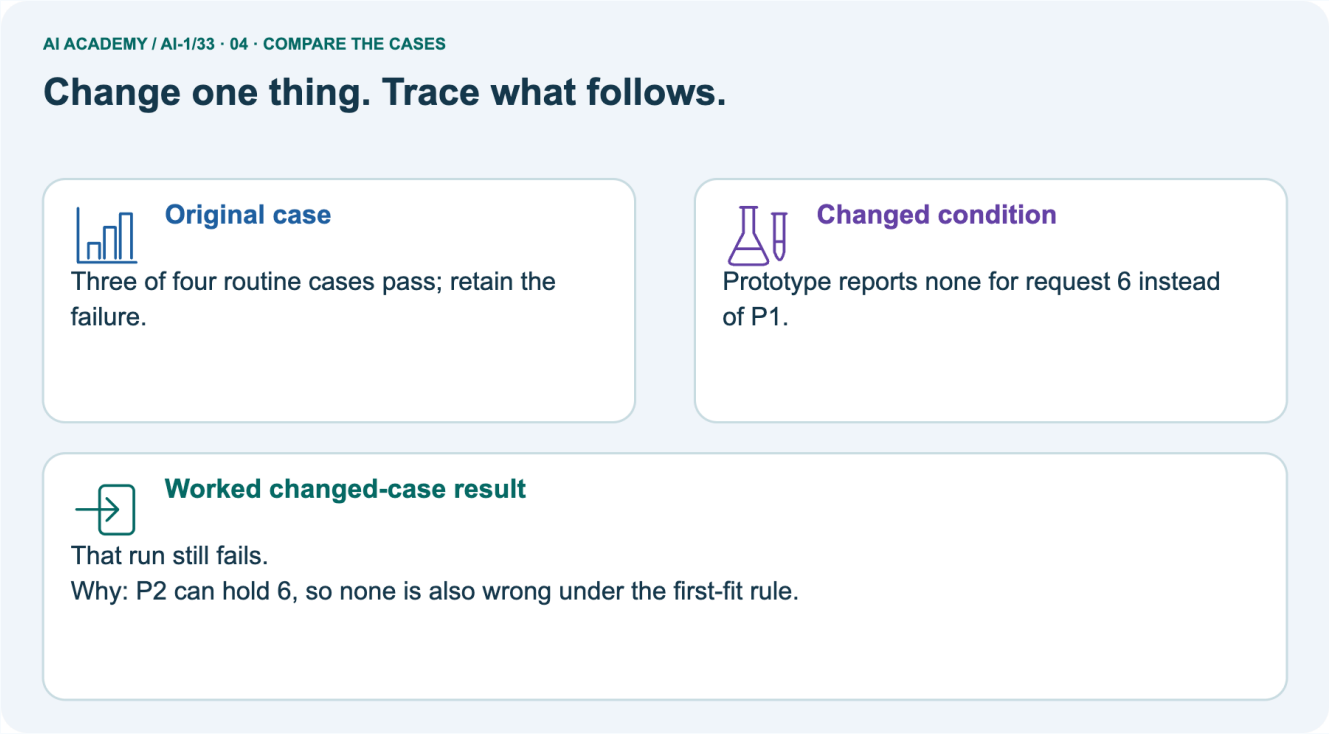
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected versus recorded routine **output**s. Test 1: Request = 2, Expected = P1, Recorded = P1, Result = pass. Test 2: Request = 4, Expected = P2, Recorded = P2, Result = pass.

Test 3: Request = 6, Expected = P2, Recorded = P1, Result = fail. Test 4: Request = 8, Expected = none, Recorded = none, Result = pass.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



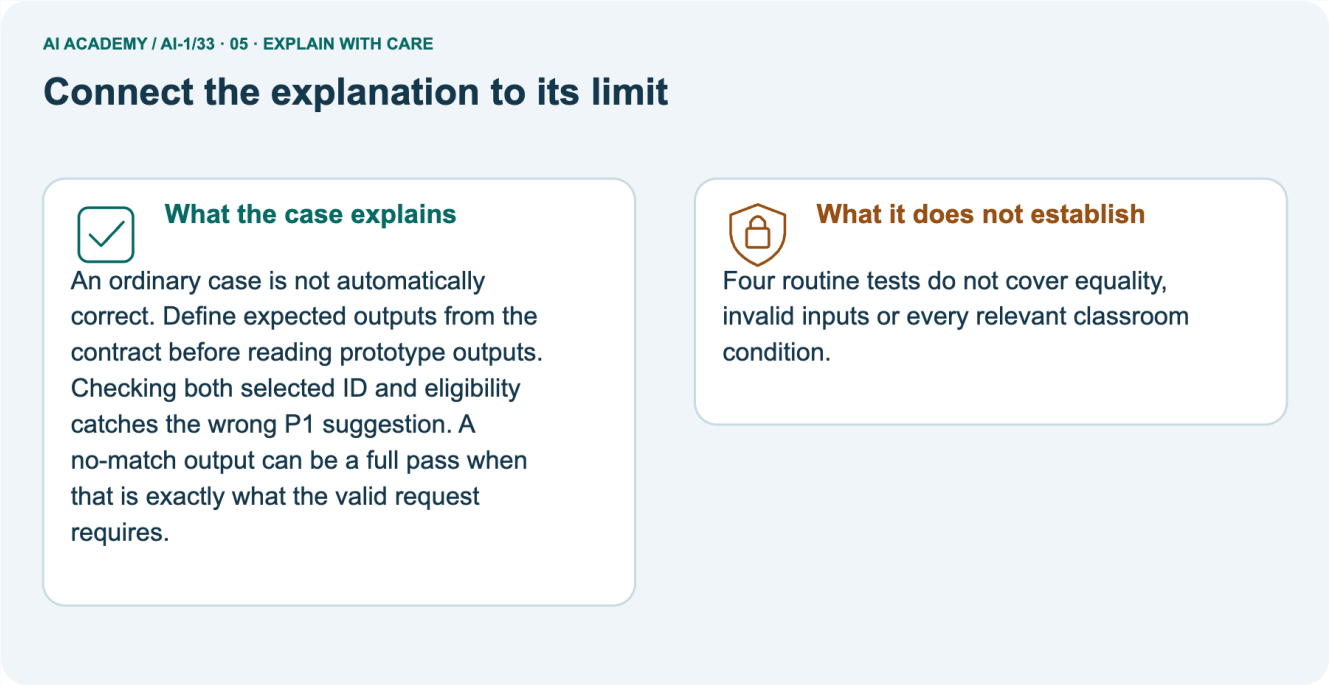
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four routine cases pass; retain the failure. Changed condition: Prototype reports none for request 6 instead of P1. **Expected result**: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



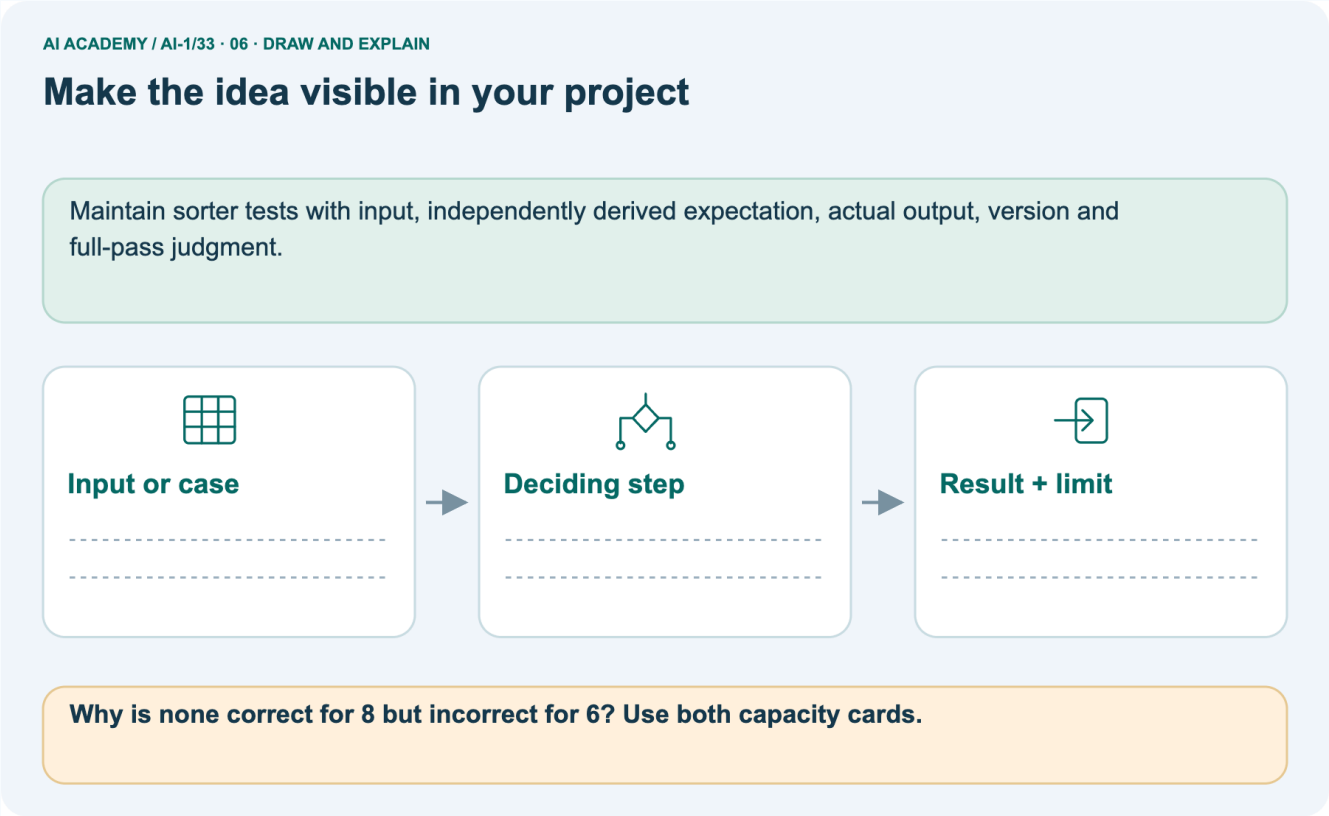
Follow the labelled picture with the matching explanation. Open the vector diagram

An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires. Limit: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment. Draw your **input**, deciding step, result and limit.

Supported question: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.